Graph — Complete DSA Sheet

All graph patterns covered | ~75 problems | Easy to Hard | LeetCode + GFG

1. GRAPH FUNDAMENTALS — Build This First

■ Represent graph as adjacency list. Know directed vs undirected, weighted vs unweighted.

Graph Representation

- ■ Build Adjacency List from Edge List
- ■ Build Adjacency Matrix
- ■ BFS Traversal of Graph
- ■ DFS Traversal of Graph
- ■ Count Number of Nodes at Each Level — BFS

2. BFS — Breadth First Search (8 problems)

■ Use Queue. Best for shortest path in unweighted graphs.

BFS Problems

- ■ Number of Islands
- ■ Flood Fill
- ■ Rotting Oranges
- ■ 01 Matrix — Distance to Nearest 0
- ■ Walls and Gates
- ■ Shortest Path in Binary Matrix
- ■ Word Ladder I (Hard) ■
- ■ Word Ladder II (Hard) ■

3. DFS — Depth First Search (8 problems)

■ Use Stack / Recursion. Best for connected components, paths, cycles.

DFS Problems

- ■ Number of Connected Components
- ■ Max Area of Island
- ■ Clone Graph
- ■ Path Exists in Graph
- ■ All Paths From Source to Target
- ■ Pacific Atlantic Water Flow
- ■ Surrounded Regions
- ■ Number of Enclaves

4. CYCLE DETECTION (6 problems)

■ Undirected: use visited + parent. Directed: use visited + recursion stack.

Cycle Detection

- ■ Detect Cycle in Undirected Graph — BFS
- ■ Detect Cycle in Undirected Graph — DFS
- ■ Detect Cycle in Directed Graph — DFS
- ■ Detect Cycle in Directed Graph — Kahn's BFS
- ■ Find Eventual Safe States
- ■ Course Schedule I — Can Finish?

5. TOPOLOGICAL SORT (5 problems)

■ Only for DAG (Directed Acyclic Graph). Two methods: DFS or Kahn's BFS.

Topo Sort

- ■ Topological Sort — DFS Method
- ■ Topological Sort — Kahn's BFS / BFS Method
- ■ Course Schedule II — Return Order

■ Alien Dictionary (Hard) ■
■ Sequence Reconstruction

6. UNION FIND / DSU (7 problems)

■ Disjoint Set Union — fastest way to check/merge connected components

Union Find

■ Implement Union Find with Path Compression + Union by Rank
■ Number of Provinces
■ Redundant Connection
■ Accounts Merge (Hard) ■
■ Making a Large Island (Hard) ■
■ Swim in Rising Water (Hard) ■
■ Number of Islands II — Dynamic

7. SHORTEST PATH (8 problems)

■ Dijkstra for non-negative weights. Bellman-Ford for negative. BFS for unweighted.

Dijkstra — Non-negative weights

■ Dijkstra Algorithm — Classic
■ Network Delay Time
■ Path with Minimum Effort
■ Cheapest Flights Within K Stops — Modified Dijkstra

Bellman Ford — Negative weights

■ Bellman Ford Algorithm — Classic
■ Cheapest Flights Within K Stops — Bellman Ford

Floyd Warshall — All pairs

■ Floyd Warshall — Classic
■ Find the City With the Smallest Number of Neighbors

8. MINIMUM SPANNING TREE — MST (4 problems)

■ Prim's or Kruskal's — connect all nodes with minimum total edge weight

MST

■ Prim's Algorithm
■ Kruskal's Algorithm
■ Min Cost to Connect All Points
■ Minimum Spanning Tree — GFG

9. BIPARTITE / GRAPH COLORING (4 problems)

■ Can graph be colored with 2 colors? Use BFS/DFS to check.

Bipartite

■ Is Graph Bipartite? — BFS
■ Is Graph Bipartite? — DFS
■ Possible Bipartition
■ Coloring a Border

10. STRONGLY CONNECTED COMPONENTS (3 problems)

■ Kosaraju's or Tarjan's — advanced, good to know for FAANG

SCC

■ Kosaraju's Algorithm — SCC
■ Tarjan's Algorithm — SCC (Hard) ■
■ Critical Connections in Network — Bridges (Hard) ■

11. MATRIX AS GRAPH (6 problems)

■ Treat each cell as a node — 4 or 8 directional movement

Grid Graph

- ■ Number of Islands
- ■ Max Area of Island
- ■ Shortest Bridge (Hard) ■
- ■ Cut Off Trees for Golf Event (Hard) ■
- ■ Minimum Steps to Reach Target (BFS on Grid)
- ■ Walls and Gates

12. ADVANCED GRAPH (6 problems)

■ Hard problems — attempt after all above are done

Hard / Advanced

- ■ Graph Valid Tree
- ■ Reconstruct Itinerary — Eulerian Path (Hard) ■
- ■ Parallel Courses II — Bitmask + BFS (Hard) ■
- ■ Bus Routes (Hard) ■
- ■ Minimum Number of Vertices to Reach All Nodes
- ■ Number of Good Paths (Hard) ■

SUMMARY

#	Graph Pattern	Problems	Hard
1	Fundamentals	5	0
2	BFS	8	2
3	DFS	8	0
4	Cycle Detection	6	0
5	Topological Sort	5	1
6	Union Find / DSU	7	3
7	Shortest Path	8	0
8	MST	4	0
9	Bipartite / Coloring	4	0
10	SCC — Kosaraju / Tarjan	3	2
11	Matrix as Graph	6	2
12	Advanced Graph	6	4
TOTAL		**70**	**14**

Recommended Order: Fundamentals → BFS → DFS → Cycle Detection → Topo Sort → Union Find → Shortest Path → MST → Bipartite → SCC → Advanced

■ Red = Hard | ■ = Not done | BFS + DFS + Union Find = 80% of graph interviews